

How the models performed

Nine Whisper configurations – three decode modes across three model sizes – run against 51.2 minutes of Sinhala–English code-switched speech and scored on word- and character-error rate, error composition, and real-time factor.

234 recording-configs decoded 26 recordings scored 7 held-out eval recordings faster-whisper / CTranslate2, float16

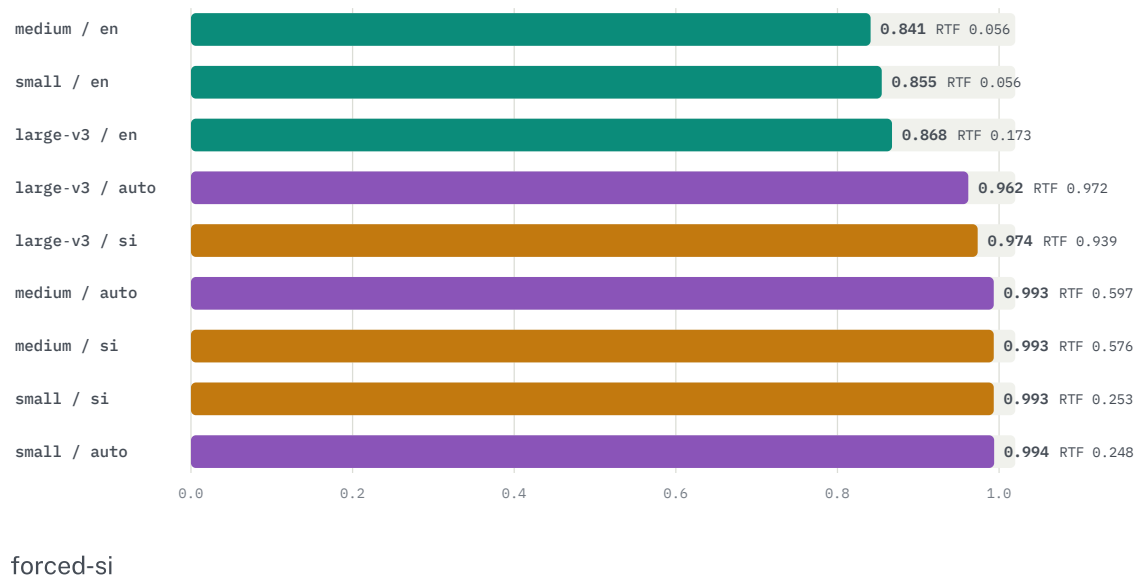
WINNING CONFIGURATION medium · forced-en lowest eval WER of all 9 configs	EVAL-SPLIT WER (FOLDED) 0.841 raw 0.847 · CER 0.588
REAL-TIME FACTOR 0.056 ≈18× faster than real time	WORST CONFIGURATION 0.994 small · auto-detect

01 The full ranking

All nine configs — {auto, forced-en, forced-si} × {small, medium, large-v3} — scored on the same 7-recording eval split, using word error rate after folding both hypothesis and reference into a common script-neutral space (§4 explains the fold). Lower is better; RTF is decode time ÷ audio duration.

Eval-split WER by configuration

Sorted best → worst. Bar color is decode mode; RTF is printed at the bar end.



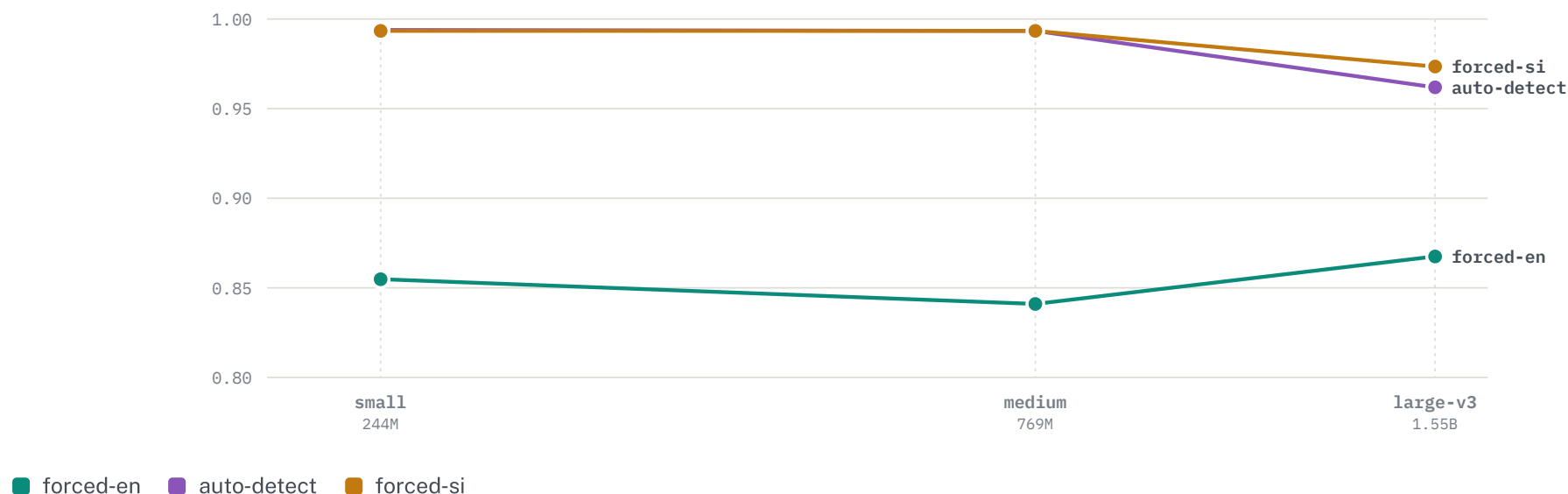
Three configs cluster at the top, all forced-English, all under 0.87 WER. Then a cliff: every non-English config lands between 0.96 and 0.99 — a gap of roughly 10 WER points that no amount of model size closes. Decode mode, not size, is what separates a working transcript from an unusable one.

02 Model size is not monotonic

§5 of the implementation plan predicted large-v3 would show the largest gain, hedged as unverified pending this test. It didn't — on the mode that actually transcribes, medium wins outright.

Eval WER vs. model size, by decode mode

X positioned by parameter count (log scale) — small 244M, medium 769M, large-v3 1.55B.



Under auto and si, size barely moves the number — both stay pinned between 0.96 and 0.99 regardless of parameter count, because the failure there isn't a capacity problem (§3). Under en, size *does* matter, but not in a straight line: WER dips at **medium** (0.841) and comes back *up* at **large-v3** (0.868) — tripling the parameters makes the transcript worse, not better.

MODEL	EVAL WER	FOLD CER	HIT	SUB	DEL	INS	RTF
small / en	0.855	0.632	0.188	0.674	0.138	0.164	0.056
medium / en	0.841	0.588	0.237	0.638	0.125	0.131	0.056
large-v3 / en	0.868	0.584	0.249	0.700	0.052	0.163	0.173

large-v3 isn't simply worse — it makes a different trade. Its deletion rate collapses to 5% (it transcribes almost everything the small/medium models leave silent) and it has the best CER of the three. But substitutions climb to 70% and insertions to 16%: it hears more and mangles more of what it hears into extra, wrong tokens. WER penalizes every one

of those insertions, and for a pipeline where downstream components (C2 normalization, C3 diarization fusion) consume individual word tokens, a correct token is what matters — so WER, not CER, is the right metric to decide on here.

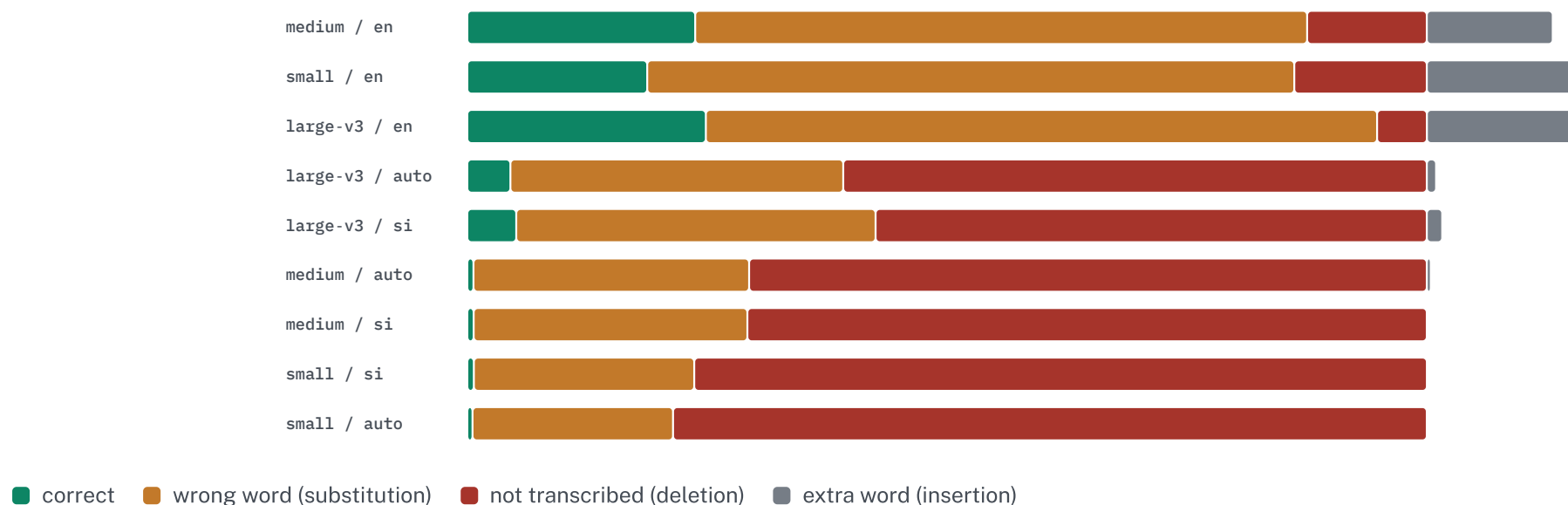
Practically convenient: the model that wins on accuracy (medium) also costs nothing over small at RTF 0.056, and both sit comfortably inside the 8 GB on-device target — large-v3 alone held ~5.7 GB resident at inference. The size-vs-deployability trade the plan anticipated doesn't arise here.

03 Where the error comes from

A WER near 1.0 can mean two very different things: the model stayed silent, or the model spoke and got it wrong. Splitting every error into substitutions, deletions, and insertions tells them apart.

Error composition, whole corpus (folded)

Share of reference words in each outcome. Same ranking order as §1.



The six lower bars are almost entirely red: 57–79% of reference words are **never produced at all**. Under auto, Whisper detects Sinhala on nearly every recording and its Sinhala decoder falls into repetition and temperature-fallback loops rather than transcribing. Forcing si reproduces the same collapse almost exactly — confirming auto and si are the same experiment, since auto-detect already lands on Sinhala.

Forcing en changes the failure mode entirely: deletions drop to 5–14%, and the dominant color becomes amber — the model is *trying*, and mostly getting individual words wrong, not refusing to speak. That's the profile fine-tuning can act on. The output is a recognisably English-phonetic rendering of code-mixed speech:

Good morning Ceylon Electricity Board Kaduwela office eken, mama Chaminda. Oyata connection ekak oneda?

gold — recording J26DS313_R0017

Good morning, Ceylon Electricity Board, Cardwell Office. Can I come in? Who's the connection?

medium / en hypothesis

English spans survive intact; Sinhala spans become plausible-sounding English. Folded CER (0.588) is far better than folded WER (0.894) for this config — the model is approximating the phonetics while missing whole word boundaries, which is exactly the gap a fine-tune on real transcripts is positioned to close.

04 The script-mismatch hypothesis was wrong

The corpus is script-split — 17 recordings write Sinhala in native Unicode, 8 romanized, 1 mixed. The working hypothesis going in was that scoring a native-script hypothesis against a romanized reference (or vice-versa) was inflating WER. It wasn't.

Folding maps both hypothesis and reference into a common romanized space before scoring — කොහොමද and kohomada both become kohomada — so any gap between *raw* and *folded* WER is purely the cost of script mismatch, with recognition quality held constant. Across all nine configs and three script groups, that cost tops out at **0.026** WER points. Essentially nothing.

CONFIG	MIXED	NATIVE	ROMANIZED
medium / en	0.004	0.008	0.000
small / en	0.008	0.010	0.002
large-v3 / en	0.004	0.007	0.002
large-v3 / auto	0.007	0.002	0.023
large-v3 / si	0.026	0.004	0.014
medium / auto	0.000	0.002	0.006
medium / si	0.000	0.002	0.002
small / si	0.007	0.003	0.006

CONFIG	MIXED	NATIVE	ROMANIZED
small / auto	0.007	0.002	0.005

Raw WER – folded WER, by script group. All values in WER points.

A useful negative result: script normalization has nothing left to offer the WER, so which script to standardize training data on can be decided on downstream convenience — not accuracy.

05 Full results table

Every config, every metric. raw scores hypothesis against reference as annotated; folded removes script as a variable (§4); RTF is decode time ÷ audio duration, measured on an RTX 5050 (8 GB).

CONFIG	RAW WER	FOLD WER	RAW CER	FOLD CER	EVAL FOLD WER	TRAIN FOLD WER	RTF
● medium / forced-en	0.900	0.894	0.651	0.588	0.841	0.918	0.0562
● small / forced-en	0.985	0.976	0.710	0.632	0.855	1.032	0.0557
● large-v3 / forced-en	0.920	0.915	0.656	0.584	0.868	0.936	0.1728
● large-v3 / auto-detect	0.971	0.965	0.835	0.740	0.962	0.966	0.9719
● large-v3 / forced-si	0.972	0.965	0.826	0.731	0.974	0.961	0.9387
● medium / auto-detect	1.001	0.998	0.891	0.804	0.993	1.000	0.5971
● medium / forced-si	0.997	0.994	0.881	0.790	0.993	0.995	0.5761
● small / forced-si	0.998	0.994	0.900	0.818	0.993	0.994	0.2528
● small / auto-detect	0.998	0.995	0.902	0.819	0.994	0.995	0.2483

Bar fill in the eval-WER column is scaled 0–1.05 across the whole table for at-a-glance comparison.

Harness check: small/auto, scored under the prior evaluation's own normalizer, reproduces its published 0.9372 micro WER within 0.0153 (0.9525 measured here) — confirming this harness reads the same data as SORA_Dataset/results/c1_report.md before any of the numbers above are trusted.

06 Per-recording breakdown – the winning config

medium / en across all 26 scorable recordings, best → worst. Script group is read from the transcript text itself, independent of any language label.

RECORDING	SCRIPT	SPLIT	RAW WER	FOLDED WER	FOLDED CER
J26DS313_R0002	● Romanized	● train	0.649	0.649	0.561
J26DS313_R0013	● Native	● train	0.712	0.705	0.441
J26DS313_R0012	● Native	● eval	0.738	0.718	0.412
J26DS313_R0018	● Mixed	● train	0.730	0.727	0.509
J26DS313_R0026	● Native	● train	0.779	0.756	0.540
J26DS313_R0019	● Native	● eval	0.789	0.776	0.576
J26DS313_R0025	● Native	● eval	0.799	0.796	0.557
J26DS313_R0014	● Romanized	● eval	0.800	0.800	0.536
J26DS313_R0024	● Romanized	● train	0.823	0.823	0.541
J26DS313_R0020	● Native	● eval	0.837	0.836	0.565
J26DS313_R0015	● Native	● train	0.849	0.848	0.582
J26DS313_R0011	● Native	● train	0.916	0.910	0.582
J26DS313_R0003	● Romanized	● train	0.914	0.914	0.540

RECORDING	SCRIPT	SPLIT	RAW WER	FOLDED WER	FOLDED CER
J26DS313_R0022	● Native	● train	0.918	0.918	0.555
J26DS313_R0027	● Native	● train	0.929	0.926	0.704
J26DS313_R0006	● Native	● train	0.991	0.944	0.608
J26DS313_R0017	● Romanized	● train	0.966	0.966	0.556
J26DS313_R0016	● Native	● train	0.992	0.979	0.606
J26DS313_R0028	● Romanized	● train	0.988	0.988	0.646
J26DS313_R0004	● Romanized	● train	1.040	1.040	0.709
J26DS313_R0005	● Native	● train	1.060	1.060	0.640
J26DS313_R0021	● Native	● train	1.097	1.083	0.569
J26DS313_R0023	● Romanized	● eval	1.121	1.121	0.648
J26DS313_R0010	● Native	● eval	1.141	1.127	0.644
J26DS313_R0009	● Native	● train	1.111	1.164	0.713
J26DS313_R0008 ★	● Native	● train	4.318	4.227	2.392

★ R0008's reference transcribes 22 words for 62.5s of audio (56% of the timeline uncovered) — its WER of 4.227 measures an incomplete annotation, not the model. Excluded from the corpus-level averages reported by the team's data-quality audit; retained here for transparency.

07 Method

How these numbers were produced, so they can be reproduced or challenged.

Corpus & split

27 recordings carry gold transcripts; one (R0007) has no matching audio file and is excluded, leaving **26 scorable recordings / 51.2 minutes**. The split is locked once, seeded, and stratified by script group so later fine-tuning

comparisons stay valid against the same held-out set:

Train	19 recordings	35.6 min
Eval	7 recordings	15.6 min
Native script	17 rec.	—
Romanized	8 rec.	—
Mixed	1 rec.	—

Decode grid

faster-whisper (CTranslate2, float16, CUDA) · beam size 5 · VAD-filtered · word-level timestamps and per-word probability captured throughout. Three decode modes — language auto-detected per segment, forced to en, forced to si — crossed with three model sizes — small (244M), medium (769M), large-v3 (1.55B) — for 9 configurations, each run against all 26 recordings: 234 decodes in total.

Scoring

WER/CER computed with `jwer`, pooled (micro) across whole-recording concatenated text — Whisper's own segmentation never aligns to gold utterance ids, so segmentation error is absorbed into the word-level score rather than discarded. Three normalizations:

- **raw** — lowercase, strip punctuation, keep whatever script the text is in. The headline figure.
- **folded** — raw, then mapped through a Sinhala ↔ Latin folding normalizer into a common comparison space (§4).
- **legacy** — bug-compatible with the prior evaluation script, which strips Sinhala vowel signs along with punctuation. Kept only to reproduce the prior baseline; never a reported number.

The decision metric is **eval-split micro WER under folded normalization** — script-agnostic, so it isolates recognition quality from a corpus quirk that fine-tuning can move (which script the model outputs) but not fix (words it never heard).

08 Reading these numbers

52 minutes, not 15–20 hours. The implementation plan's §6.2 assumes roughly 1 hour held out for evaluation plus 15–20 hours for training. The eval split alone accounts for most of the currently available audio. Every number on this page is a Phase 0 baseline, not a converged result — none of these models have been fine-tuned yet.

Two references are incomplete. R0008 and R0005 have gold transcripts that cover well under half their audio's timeline, inflating their WER independent of model quality (R0008 is flagged in §6). Both should be excluded from ASR scoring pending re-annotation.

Folded WER is a floor, not a headline. The folding normalizer in §4 is deliberately many-to-one and will merge genuinely distinct words that happen to romanize the same way. It's the right tool for isolating script-mismatch cost; it is not the number to quote as final accuracy.

A separate data-quality audit — token-level language labels, the OTHER/Tamil class, two systematically mislabelled recordings — feeds Component 1's language-ID work rather than the ASR numbers on this page, and is documented alongside the corpus (`docs/TOKEN_SCHEMA.md`).

Machine-readable results: `eval/baseline_decisions.json` . Reproduce with:

```
python -m src.decode_compare  
python -m src.evaluate_asr
```

Component 1 · J26-DS-313 · Phase 0, Week 0 (Aug 30 – Sep 5 2026) · faster-whisper on an RTX 5050 (8 GB)